Session 5: Exoplanets from Light Curves

Transit geometry, detrending, the box-least-squares periodogram

astro-lab – reading notes

Status: reading notes and exercise specification. The code for this session (`05_lightcurves/`) is not in the repository yet; the exercises below define what to build.

Goals

Predict the depth and duration of a transit from physical parameters, then recover a planet's period from a noisy light curve with a box-least-squares search, first on synthetic data and then on real Kepler or TESS data via Lightkurve.

1 Transit geometry

A planet of radius R_p crossing a star of radius $R_\star$ blocks a fraction of light

$$\delta \approx \left(\frac{R_p}{R_\star}\right)^2 = k^2.$$

Examples: a Jupiter-sized planet crossing a Sun-like star gives $\delta \approx 1\%$; an Earth-sized planet gives $\delta \approx 84$ ppm.

For a circular orbit with impact parameter b the total duration (first to fourth contact) is approximately

$$T_{14} \approx \frac{P}{\pi}\frac{R_\star}{a}\sqrt{(1+k)^2 - b^2},$$

valid for $a \gg R_\star$. The semi-major axis follows from Kepler's third law (Session 1), $a^3 = GM_\star P^2/4\pi^2$. For Earth around the Sun this gives $T_{14} \approx 13$ hours for $b = 0$.

> **Detection is a signal-to-noise problem.** With per-point noise σ and N_{in} in-transit points, $\mathrm{SNR} \approx \delta\sqrt{N_{\mathrm{in}}}/\sigma$. Small planets need many transits and long baselines.

2 Preparing a light curve

1. Remove bad-quality cadences and normalise by the median flux.
2. **Flatten** slow stellar and instrumental trends with a smoothing filter whose window is several times the expected transit duration; a window that is too short will erase the transit itself.
3. Optionally clip upward outliers (flares), never downward ones.

3 Box-least-squares (BLS)

BLS models the light curve as a constant level with a periodic rectangular dip, fitted over a grid of trial periods P, durations τ and epochs t_0. At each trial period the data are phase-folded; for each (τ, t_0) the in-box mean is compared with the out-of-box mean, and the statistic reported is the reduction in χ^2 (or SNR) relative to a flat model. The best peak in the periodogram gives P, τ, t_0 and a depth.

Practical points:

- The period grid must be fine enough that phase drift over the baseline T is smaller than the duration: $\Delta P \lesssim P\,\tau/T$.
- Peaks also appear at $P/2$, $2P$, $P/3$; check by folding at each.
- A single deep dip does not give a period; you need at least two transits.

4 Implementation plan

1. Simulate: flat light curve, 2-minute cadence over 27 days (one TESS sector), inject a box transit with $P = 3.7$ d, $\delta = 0.5\%$, $\tau = 2.5$ h, add Gaussian noise of 0.1%.
2. Search with `astropy.timeseries.BoxLeastSquares`; confirm it recovers P within the grid resolution.
3. Reduce the injected depth until recovery fails; plot recovery rate versus SNR.
4. Repeat on a real target with Lightkurve: download a light curve, flatten it, compute a BLS periodogram with `to_periodogram`, and fold at the best period.

5 Exercises

1. Verify the Earth and Jupiter depths and compute T_{14} for both around the Sun.
2. Show that the transit probability for a random orientation is $\approx R_\star/a$ and evaluate it for $P = 3.7$ d and $P = 365$ d.
3. Inject the transit of a hot Jupiter into a real, quiet star's light curve and test whether your pipeline recovers it.
4. Compare BLS with a Lomb–Scargle periodogram on the same data and explain why Lomb–Scargle is poorly suited to box-shaped dips.

References

- Winn (2010), *Transits and Occultations*, in *Exoplanets* (arXiv:1001.2010)
- Kovács, Zucker, Mazeh (2002), *A&A* 391, 369 (BLS)
- Lightkurve documentation: https://lightkurve.github.io/lightkurve/